COLLEGE EQUIPMENT RESOURCE MANAGER

A PROJECT REPORT

Submitted by

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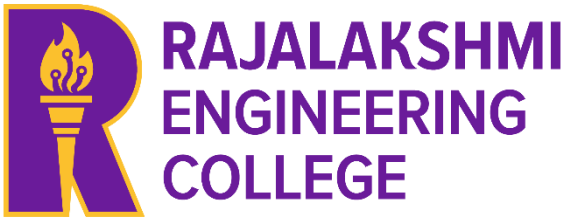
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CB23F35 –
LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

In

BACHELOR OF ENGINEERING

COMPUTER SCIENCE AND ENGINEERING



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

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[OCTOBER, 2026]

RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "**College Equipment Resource Manager**" is the bonafide work of **CHEMBETI JAHNAVI (240701089)**, who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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DECLARATION

I, CHEMBETI JAHNAVI (240701089), hereby declare that the project report titled "College Equipment Resource Manager", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report was developed by me, and the report describes it from the application's exported definition and screenshots.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report. The structure of this report follows a sample report supplied by the course faculty as a format reference; none of its project content has been reused. The use of Zia or any other AI assistance is stated in Section 8.6.

Place: Chennai
Date: 08 November 2026

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CHEMBETI JAHNAVI

ABSTRACT

Colleges own a large amount of laboratory and workshop equipment, yet its borrowing, approval, return, damage and repair are commonly handled through registers, spreadsheets and informal messages. These records are scattered and hard to search, and they give no clear picture of which items are available, who holds them, when they are due and what has been spent on repairs. This project presents the College Equipment Resource Manager, an application developed on the Zoho Creator low-code platform to keep these records in one place. Low-code development was chosen because the platform supplies the database, forms, reports, pages, workflows and hosting in a single environment, so the effort goes into the equipment process itself. The application has seven forms (Department Registration, Equipment Registration, Borrow Request, Request Approval, Equipment Issue Return, Damage Report and Maintenance Request) with 75 fields joined by eleven lookups. Thirteen reports, seven lists and six status or type boards, present the records, and a Dashboard shows total purchase and repair costs, the number of issue and return records, the condition of equipment at return and repair cost by maintenance type and status. Three Deluge scripts validate the dates of a borrow request, e-mail the department when a request is urgent and remind departments of warranty expiry, and two Blueprints guide a borrow request through 13 stages and a maintenance request through 14 stages. The report describes the analysis, data model, reports, workflows, architecture, security and test plan of the application. A review of the exported definition identified points for correction, including role-specific profiles, mandatory fields, an inactive reminder schedule and a notification that tests only the Urgent priority. The test cases are defined and are to be recorded at the demonstration, and the application is not deployed to college users. The work shows that a small, connected equipment-management system can be built largely by configuration.

Keywords: Low-Code Development; Zoho Creator; Equipment Management; Resource Management; Workflow Automation; Deluge Scripting

PROJECT AT A GLANCE

Attribute	Details
Project Title	College Equipment Resource Manager
Application Domain	Education – college equipment and laboratory resource management
Student Name & Register No.	CHEMBETI JAHNAVI – 240701089
Course Faculty	Ms. Aswana Lal
Zoho Creator Edition	Zoho Creator trial edition (trial banner visible in the screenshots)
Application Link Name	college-equipment-resource-manager (account jahnavichembeti)
Published / Live URL	https://creatorapp.zoho.in/jahnavichembeti/college-equipment-resource-manager (live mode; not published to college users)
Demo Credentials (per role)	Not included in this report
Number of Forms / Subforms	7 forms / 0 subforms; 75 fields; 11 lookups
Number of Reports (by type)	13 reports: 7 list and 6 kanban
Number of Pages / Dashboards	1 page (Dashboard)
Number of Workflows	5: 2 form workflows, 1 schedule workflow (inactive) and 2 Blueprints; no approvals or batch workflows
Deluge Functions / Custom Actions	3 workflow scripts (Appendix B); no standalone functions
Integrations	E-mail through sendmail only
Roles & Profiles Defined	Default profiles only (Read, Write, Developer, Administrator, Customer); default role CEO
Environments Used	Single application (live mode)
Date of Production Deployment	Not deployed to college users

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
Deluge	Zoho's scripting language (Data Enriched Language for the Universal Grid Environment)
ER	Entity–Relationship
FR	Functional Requirement
KPI	Key Performance Indicator
NFR	Non-Functional Requirement
RBAC	Role-Based Access Control
SDLC	Software Development Life Cycle
TC	Test Case
UAT	User Acceptance Testing
UI	User Interface
URL	Uniform Resource Locator
Zia	Zoho's AI assistant

CHAPTER 1
INTRODUCTION

1.1 Introduction to College Equipment Resource Management

A college owns equipment of many kinds: oscilloscopes, computers, measuring instruments, mechanical tools, electrical machines and laboratory apparatus. Each department buys equipment for its own courses and keeps it in its own laboratories. The equipment is a significant investment, and it needs to be registered, issued, returned, inspected, repaired and, in time, retired.

Resource management of this kind is more than keeping a list of items. It covers the identity of each item, the department that owns it, whether it is available, the request made by someone who wants to use it, the approval of that request, the movement of the item from one person to another, its condition when it comes back, any damage, and the repair that follows. When these details are kept in separate registers and spreadsheets, a simple question such as “which items are with borrowers today?” or “how much have repairs cost this year?” takes much effort to answer. A light and easily maintained system that connects all these records is therefore useful for a college.

1.2 Problem Statement

Colleges lack a simple, connected system that follows equipment from registration to borrowing, return, damage and repair. As a result, staff must search several registers and spreadsheets to learn what is available, who holds an item, whether a request was approved, and what repairs have cost. The following pain points are typical:

- **Scattered records:** equipment lists, borrowing entries and repair notes are kept in different places, which leads to duplicate and inconsistent entries.
- **Unclear availability:** a person who needs an item cannot easily see whether it is free, borrowed or under repair.
- **Informal approval:** requests made by phone or message are easily forgotten, and no record shows who approved what and when.
- **Weak return tracking:** the expected return date is remembered rather than recorded, and the condition of the item at return is often not noted.
- **Poor cost visibility:** purchase costs and repair costs are not brought together, so the cost of keeping equipment working is not seen.

1.3 Project Objectives

- To design a connected data model and forms for departments, equipment, borrow requests, approvals, issue and return, damage reports and maintenance requests.

- To provide reports that list the records of each form and summarise them by status, type and category.
- To provide a Dashboard that shows cost figures, record counts and the condition of equipment at return.
- To automate selected steps through workflows and Blueprints, including validation of request dates, notification about urgent requests, a reminder before warranty expiry and defined stages for borrowing and maintenance.
- To define role-based access so that each user sees only the information needed for his or her work; this objective is only partly met, because the application has default profiles only (Chapter 12).
- To record clearly which features are implemented and verified, which are under development, and which are planned for the future.

1.4 Scope of the Project

In scope. The application covers department registration; equipment registration with category, cost and warranty details; borrow requests with a priority; approval of requests; recording of equipment issue and return with dates and condition; damage reports; maintenance requests with type, status and repair cost; reports; and a Dashboard. The platform is Zoho Creator, accessed through a web browser.

Out of scope. The application does not track the physical location of equipment in real time, does not read data from sensors or barcodes, does not handle purchasing or depreciation accounting, and does not connect to the college's examination or administration systems.

1.5 Existing System

In a typical college, the existing system is manual or semi-manual. Equipment is entered in a stock register or a spreadsheet, usually maintained by one laboratory in-charge. A department that needs an item from another department asks by phone, message or a written note. The reply is informal, and the issue and return are written in a register, if at all. Expected return dates depend on memory. Damage is reported verbally, and repair details and costs are kept apart from the borrowing record. Any report, such as a list of items currently borrowed, has to be compiled by hand.

1.6 Proposed System

The proposed system is the College Equipment Resource Manager, built on Zoho Creator. It replaces the registers and spreadsheets with seven connected forms, reports for the records, a Dashboard, and workflows for validation, notification and reminders. A borrow request is entered once and then followed through approval, issue, return,

damage and maintenance in linked forms. The features of the application are described in Chapters 3 to 6, and the planned features that were not built are labelled as future work.

1.7 Advantages of the Proposed System

- **Centralised information:** equipment, borrowing and repair records are stored in one database.
- **Visibility:** reports and the Dashboard show the state of the equipment without manual compilation.
- **Controlled requests:** borrow requests carry a priority and are validated by a workflow before they are accepted.
- **Cost awareness:** purchase cost and repair cost are shown on the Dashboard.
- **Condition tracking:** the condition of equipment at return is recorded and summarised.
- **Low cost and fast change:** forms, reports and workflows can be modified without extensive programming.

1.8 Justification for a Low-Code Approach

A low-code platform is suitable for this problem because the data model is relational but small, the business rules are few, and the user base is limited to one institution. Zoho Creator provides form building, lookup relationships, reports, pages, workflows, the Deluge scripting language, hosting and security in one environment [1], [2]. Building the same application from scratch would need a database design, a web interface, hosting and authentication, and the effort would be spent on infrastructure instead of on the equipment process. Low-code platforms are also described as allowing applications to be built mainly through configuration with limited hand-written code [3], [4].

1.9 Basis of the Report and Implementation Status

This report has been prepared from three sources: the exported definition of the application (a .ds file generated on 08-Nov-2026), four screenshots of the application, and the project details supplied by the student. Forms, fields, choices, lookups, reports, the Dashboard, workflows, Blueprints and Deluge scripts are taken from the export. Behaviour that can only be seen by running the application, such as the response of a form to a user’s entry, is not claimed as tested. The status terms are used as follows.

- **Found:** the item is present in the exported definition or in a screenshot. This shows that it exists, not that it has been tested.

- **Test case defined:** the expected behaviour is written in Chapter 11 and is to be recorded during the demonstration.
- **Future / Not built:** the item is a design intention or a recommendation and is not part of the application.

Table 1.1: Verification of the Application against the Planned Design

Area	Finding	Status
Application	College Equipment Resource Manager; live mode on the Zoho Creator account	Found
Forms	Seven forms with 75 fields and 11 lookups; no subform	Found
Reports	13 reports: 7 list and 6 kanban	Found
Dashboard	Four tiles, two bar charts, Condition at Return panel, quick links and embedded report	Found
Workflows	Two form workflows, one schedule workflow (inactive) and two Blueprints	Found
Deluge scripts	Three scripts in the workflows (Appendix B)	Found
Roles and permissions	Default profiles only; planned roles not created	Partly met
Integrations	E-mail only	Found
AI features (Zia)	No AI feature in the application	Not used
Testing	Test cases defined; results recorded at the demonstration	To be executed
Deployment	Not published to college users	Not deployed

1.10 Organisation of the Report

Chapter 2 presents the system analysis and requirements. Chapter 3 describes data modelling and forms. Chapter 4 covers reports and the Dashboard, and Chapter 5 the pages and navigation. Chapter 6 explains workflows and automation. Chapter 7 discusses integration capabilities, and Chapter 8 analytics and AI. Chapter 9 gives the system architecture, Chapter 10 the implementation sequence and Chapter 11 testing. Chapters 12 to 15 discuss security, advantages and limitations, future enhancements and the conclusion. Chapter 16 lists the references, and the appendices give an implementation evidence checklist, the Deluge source code, a screenshot checklist.

CHAPTER 2
SYSTEM ANALYSIS AND REQUIREMENTS

2.1 User Requirements

The people who use or are affected by the system are listed in Table 2.1. The roles are those intended in the project brief. The application itself has only the default Zoho Creator profiles (Chapter 12), so these roles describe how the application is meant to be used; they are not yet separate profiles.

Table 2.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Super Admin	Administrator of the whole application	Manage users and settings; view all records, reports and the Dashboard
Department Manager	Head or authorised person of a department	Approve or reject borrow requests for the department's equipment; view department information
Faculty	Teaching staff who need equipment	Submit borrow requests; track their status; report damage
Lab Technician	Staff in charge of a laboratory	Register equipment; issue and receive equipment; record condition; handle maintenance
Student	Student who needs equipment for a course	View available equipment and submit or track requests

The functional requirements are listed in Table 2.2. Each carries an identifier that is reused in the test cases of Chapter 11. The status column states what was found in the application; it does not state that the requirement has passed testing.

Table 2.2: Functional Requirements

FR ID	Requirement	Status
FR-01	The system shall register a department with its contact details and status.	Found: Department Registration form
FR-02	The system shall register equipment with its category, owning department, cost, condition, availability and warranty expiry.	Found: Equipment Registration form (14 fields)
FR-03	The system shall record a borrow request with the requesting department, period, purpose and priority.	Found: Borrow Request form with Priority
FR-04	The system shall validate the dates of a borrow request and reject invalid dates.	Found: Validate Borrow Request Dates
FR-05	The system shall notify the manager when a high-priority request is submitted.	Found: notification workflow (tests Urgent only)

FR ID	Requirement	Status
FR-06	The system shall record the approval or rejection of a borrow request.	Found: Request Approval form
FR-07	The system shall record the issue and return of equipment, with issue, expected return and actual return dates.	Found: Equipment Issue Return form
FR-08	The system shall record the condition of equipment at return.	Found: Condition at Return field and Dashboard panel
FR-09	The system shall record damage to equipment with its type.	Found: Damage Report form
FR-10	The system shall record maintenance requests with type, status and repair cost.	Found: Maintenance Request form
FR-11	The system shall follow the maintenance lifecycle of equipment.	Found: Equipment Maintenance Lifecycle Blueprint
FR-12	The system shall follow the equipment borrowing lifecycle.	Found: Equipment Borrowing Workflow Blueprint
FR-13	The system shall remind users before the warranty of equipment expires.	Found: schedule workflow (inactive)
FR-14	The system shall provide reports for the records of each form and for statuses, types and categories.	Found: 13 reports
FR-15	The system shall provide a Dashboard with cost figures, record counts and charts.	Found: Dashboard
FR-16	The system shall restrict access to forms and reports according to the user's role.	Partly met: default profiles only

The non-functional requirements are listed in Table 2.3. No performance was measured, so no figure is stated.

Table 2.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Usability	Forms and menus should use clear labels so that a new user can add a record without training.
NFR-02	Security	Access to records should depend on the user's profile.
NFR-03	Performance	Forms and reports should respond in a reasonable time for a college-sized data set.
NFR-04	Availability	The application should be available whenever the Zoho Creator service is available, with no server to be maintained by the college.
NFR-05	Maintainability	Forms, reports and workflows should be changeable through the builder without rewriting the application.
NFR-06	Data quality	Choice fields (status, condition, type) should be used in place of free text where a fixed list is suitable.

2.2 Software Requirements

Table 2.4: Software Requirements

Item	Requirement
Platform	Zoho Creator (trial edition)
Scripting language	Deluge
Browser	A current version of a web browser such as Google Chrome
Network	Internet connection
Other Zoho services	None
Mobile app	Not required; the web interface is used

2.3 Hardware Requirements

A computer or laptop able to run a current web browser with an internet connection is sufficient. No server hardware is needed because the application is hosted by Zoho.

2.4 System Feasibility

Technical feasibility. The required features (forms, lookups, reports, dashboard and workflows) are standard capabilities of Zoho Creator. **Economic feasibility.** Development used a trial edition, so no cost was incurred during the project; a long-term deployment would need a suitable paid or institutional plan; this should be checked against the plan chosen. **Operational feasibility.** Staff already keep equipment records in registers and spreadsheets, so a form-based application matches existing practice and needs little training.

2.5 Limitations of the Existing Manual System

Table 2.5 lists the limitations of the manual system together with the way the application responds to each. The right-hand column describes the intended response, and the status of the application is given in Table 1.1.

Table 2.5: Limitations of the Manual System and the Zoho Creator Response

Limitation	Response in the Application
Records scattered across registers and sheets	Seven connected forms in one database
Availability not visible	Availability Status field on the equipment record, shown in All Equipment Registrations
Informal, unrecorded approval	Request Approval form
Return dates remembered, not recorded	Issue and return dates stored in Equipment Issue Return

Limitation	Response in the Application
Damage and repair kept apart	Damage Report and Maintenance Request forms, with repair cost on the Dashboard
Costs not combined	Purchase and repair cost tiles on the Dashboard
Manual reports	Reports and charts generated from stored records

2.6 How Zoho Creator Addresses these Limitations

Zoho Creator stores the data of every form in a built-in database, links forms through lookup fields, produces reports from the stored data, runs workflows when records are saved or on a schedule, and controls access through profiles [1]. Deluge scripts add logic where configuration is not enough [2]. These capabilities allow the equipment process to be built without separate database, hosting or authentication work.

CHAPTER 3
DATA MODELLING AND FORMS

3.1 Overview of the Data Model

The data model has seven forms and 75 fields. Two forms hold master data (Department Registration and Equipment Registration). Three forms follow a borrowing from request to return (Borrow Request, Request Approval and Equipment Issue Return). Two forms hold the condition and repair history of equipment (Damage Report and Maintenance Request). Eleven lookup fields join the forms, so that a record refers to a record of another form and does not repeat its details.

3.2 Form Creation

The forms, fields, choices and lookups below are taken from the application's exported definition (the .ds file generated on 08-Nov-2026). The builder search view in Figure 3.1 shows the forms, reports, workflows and page of the application together. Every form has the standard Submit and Reset buttons on add and Update and Cancel buttons on edit, and a success message.

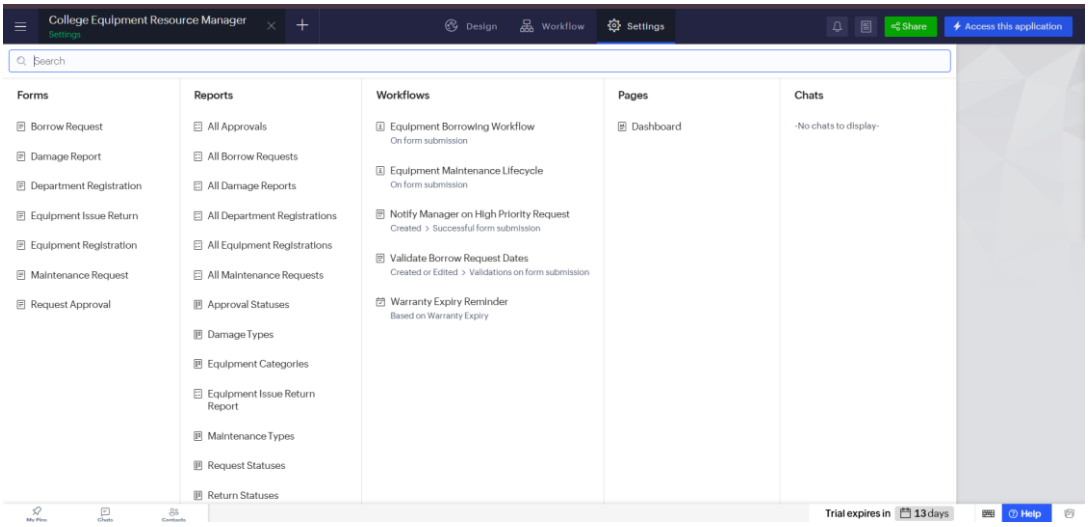


Figure 3.1: Builder Search View Listing the Forms, Reports, Workflows and Page of the Application

3.3 Summary of the Forms

Table 3.1: Forms of the Application

Form	Fields	Lookups	Success message
Borrow Request	10	1	Borrow Request Added Successfully
Damage Report	11	2	Damage Report Added Successfully
Department Registration	8	0	Department Registration Added Successfully

Form	Fields	Lookups	Success message
Equipment Issue Return	13	3	Data Added Successfully!
Equipment Registration	14	1	Equipment Registration Added Successfully
Maintenance Request	12	2	Maintenance Request Added Successfully
Request Approval	7	1	Approval Requested Successfully

3.4 Department Registration Form

Department Registration is the master list of departments. It is the parent form of three lookups: Equipment Registration, Borrow Request and Equipment Issue Return refer to it. Its Contact Email field is used by the high-priority notification workflow and its Department Head field by the warranty reminder workflow.

Table 3.2: Field Specification – Department Registration Form

Field Label	Link Name	Field Type	Details
Department ID	department_id	Single Line	
Department Name	department_name	Single Line	
Department Head	department_head	Single Line	
Contact Email	contact_email	Email	
Contact Number	contact_number	Phone	
Building	building	Single Line	
Room Lab Number	room_number	Single Line	
Department Status	department_status	Radio Buttons	Choices: Active / Inactive

3.5 Equipment Registration Form

Equipment Registration records every item that can be borrowed, with its category, owning department, cost, condition, availability and warranty expiry. The Dashboard's purchase-cost tile and the warranty reminder workflow use fields of this form. Its report shows a total, average, minimum and maximum for Purchase Cost.

Table 3.3: Field Specification – Equipment Registration Form

Field Label	Link Name	Field Type	Details
Equipment ID	equipment_id	Single Line	
Equipment Name	equipment_name	Single Line	

Field Label	Link Name	Field Type	Details
Equipment Category	equipment_category	Drop Down	Choices: Measuring Instrument / Electrical Equipment / Electronics / Computer Equipment / Mechanical Equipment / Laboratory Equipment / Other
Department	department	Lookup	Lookup to Department Registration; shows Department ID
Laboratory	laboratory	Single Line	
Manufacturer	manufacturer	Single Line	
Model Number	model_number	Single Line	
Serial Number	serial_number	Single Line	
Purchase Date	purchase_date	Date	
Purchase Cost	purchase_cost	Currency (USD)	Report shows total, average, minimum and maximum where enabled
Equipment Condition	equipment_condition	Drop Down	Choices: New / Fair / Under Maintenance / Good / Damaged
Availability Status	availability_status	Drop Down	Choices: Borrowed / Retired / Available / Under Maintenance / Reserved
Location	location	Single Line	
Warranty Expiry	warranty_expiry	Date	

3.6 Borrow Request Form

Borrow Request lets a user ask for equipment for a period and a purpose. The form carries a validation workflow, a notification workflow and the Equipment Borrowing Workflow Blueprint. The form has no subform: it does not name the equipment requested; the item is linked later, in Equipment Issue Return.

Table 3.4: Field Specification – Borrow Request Form

Field Label	Link Name	Field Type	Details
Request ID	request_id	Single Line	
Requesting Department	requesting_department	Lookup	Lookup to Department Registration; shows Department ID
Requester Name	requester_name	Single Line	
Requester Email	requester_email	Email	
Purpose	purpose	Multi Line	
Required From	required_from	Date-Time	

Field Label	Link Name	Field Type	Details
Required Until	required_until	Date-Time	
Priority	priority	Drop Down	Choices: High / Urgent / Low / Normal
Request Status	request_status	Drop Down	Choices: Approved / Completed / Rejected / Cancelled / Pending
Additional Notes	additional_notes	Multi Line	

3.7 Request Approval Form

Request Approval records the decision on a borrow request, with the date, the rejection reason and the manager's comments. Its report has a filter on Approval Date and highlights Approved and Rejected rows.

Table 3.5: Field Specification – Request Approval Form

Field Label	Link Name	Field Type	Details
Approval ID	approval_id	Single Line	
Borrow Request	borrow_request	Lookup	Lookup to Borrow Request; shows Request ID
Approving Manager	approving_manager	Single Line	
Approval Status	approval_status	Drop Down	Choices: Approved / Rejected / Pending
Approval Date	approval_date	Date	
Rejection Reason	rejection_reason	Multi Line	
Manager Comments	manager_comments	Multi Line	

3.8 Equipment Issue Return Form

Equipment Issue Return records the movement of an equipment item to a borrower and back, with the condition at issue and at return. Its report is embedded in the Dashboard and its status field is used by the Return Statuses board.

Table 3.6: Field Specification – Equipment Issue Return Form

Field Label	Link Name	Field Type	Details
Transaction ID	transaction_id	Single Line	
Borrow Request	borrow_request	Lookup	Lookup to Borrow Request; shows Request ID
Equipment	equipment	Lookup	Lookup to Equipment Registration; shows Equipment ID

Field Label	Link Name	Field Type	Details
Borrower	borrower	Single Line	
Issuing Department	issuing_department	Lookup	Lookup to Department Registration; shows Department ID
Issue Date	issue_date	Date	
Expected Return Date	expected_return_date	Date	
Actual Return Date	actual_return_date	Date	
Condition at Issue	condition_at_issue	Drop Down	Choices: New / Fair / Good / Damaged
Condition at Return	condition_at_return	Drop Down	Choices: New / Fair / Good / Damaged
Issued By	issued_by	Single Line	
Returned To	returned_to	Single Line	
Return Status	return_status	Drop Down	Choices: Not Issued / Returned / Issued / Overdue / Damaged

3.9 Damage Report Form

Damage Report records damage found on equipment, with type, severity, estimated cost and status. It is linked to the equipment and to the transaction in which the damage was found.

Table 3.7: Field Specification – Damage Report Form

Field Label	Link Name	Field Type	Details
Damage ID	damage_id	Single Line	
Equipment	equipment	Lookup	Lookup to Equipment Registration; shows Equipment ID
Transaction	transaction	Lookup	Lookup to Equipment Issue Return; shows Transaction ID
Reported By	reported_by	Single Line	
Damage Date	damage_date	Date	
Damage Type	damage_type	Drop Down	Choices: Software Issue / Electrical Failure / Missing Component / Physical Damage / Other
Damage Description	damage_description	Multi Line	
Damage Severity	damage_severity	Drop Down	Choices: Major / Moderate / Critical / Minor
Estimated Repair Cost	estimated_repair_cost	Currency (USD)	Report shows total, average, minimum and maximum where enabled
Evidence	evidence	Single Line	

Field Label	Link Name	Field Type	Details
Damage Status	damage_status	Drop Down	Choices: Under Review / Reported / Closed / Resolved / Sent for Repair

3.10 Maintenance Request Form

Maintenance Request records repair or upkeep of equipment with its type, technician, dates, cost and status. The Equipment Maintenance Lifecycle Blueprint is attached to this form, and the Dashboard's repair-cost tile and charts are built from it.

Table 3.8: Field Specification – Maintenance Request Form

Field Label	Link Name	Field Type	Details
Maintenance ID	maintenance_id	Single Line	
Equipment	equipment	Lookup	Lookup to Equipment Registration; shows Equipment ID
Damage Report	damage_report	Lookup	Lookup to Damage Report; shows Damage ID
Maintenance Type	maintenance_type	Drop Down	Choices: Preventive Maintenance / Inspection / Calibration / Repair / Other
Technician	technician	Single Line	
Request Date	request_date	Date	
Start Date	start_date	Date	
Expected Completion Date	expected_completion_date	Date	
Actual Completion Date	actual_completion_date	Date	
Repair Cost	repair_cost	Currency (USD)	Report shows total, average, minimum and maximum where enabled
Maintenance Status	maintenance_status	Drop Down	Choices: Requested / Assigned / Completed / Under Repair / Cancelled
Remarks	remarks	Multi Line	

3.11 Field Types and Validation

The forms use these field types: Single Line (identifiers and names), Multi Line (descriptions and comments), Email, Phone, Date, Date-Time (the required-from and required-until fields of a borrow request), Currency in US dollars (purchase cost, estimated repair cost and repair cost), Drop Down (fixed choices, each with an "Other" option) and Radio Buttons (Department Status). Lookups are drop-down fields filled from another form, with an "Add New" option.

The identifier fields (Request ID, Approval ID, Transaction ID, Equipment ID, Damage ID, Maintenance ID and Department ID) are Single Line text, so each identifier is typed by the user and is not generated. The exported definition contains no mandatory-field, unique-value or default-value setting, and no field-level validation. The only validation found is the date check of the **Validate Borrow Request Dates** workflow (Section 6.4).

3.12 Lookup Relationships

Table 3.9 lists the eleven lookups in the exported definition.

Table 3.9: Lookup Relationships in the Application

Form (child)	Lookup field	Refers to (parent)	Value displayed
Borrow Request	Requesting Department	Department Registration	Department ID
Damage Report	Equipment	Equipment Registration	Equipment ID
Damage Report	Transaction	Equipment Issue Return	Transaction ID
Equipment Issue Return	Borrow Request	Borrow Request	Request ID
Equipment Issue Return	Equipment	Equipment Registration	Equipment ID
Equipment Issue Return	Issuing Department	Department Registration	Department ID
Equipment Registration	Department	Department Registration	Department ID
Maintenance Request	Equipment	Equipment Registration	Equipment ID
Maintenance Request	Damage Report	Damage Report	Damage ID
Request Approval	Borrow Request	Borrow Request	Request ID

3.13 Database Relationship Design

Figure 3.2 is drawn from the lookups of Table 3.9. One department owns many equipment items and may raise many borrow requests. A request may have an approval record and one or more issue and return records. An equipment item may appear in many issue and return, damage and maintenance records. A damage report points to the transaction in which the damage was found, and a maintenance request may point to a damage report.

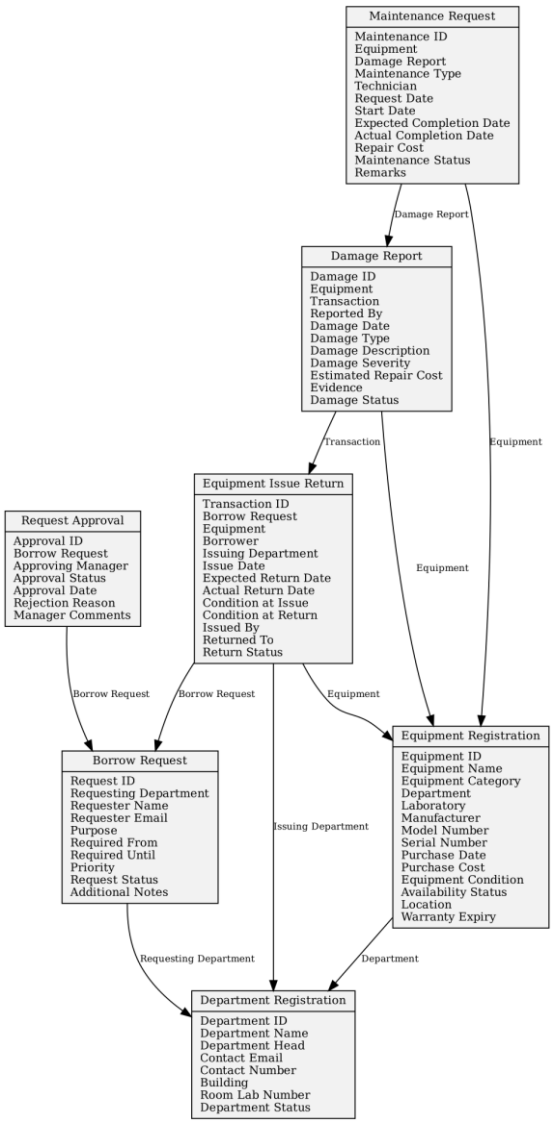


Figure 3.2: Entity–Relationship Diagram of the Seven Forms (Drawn from the Application's Lookups)

3.14 Record Management and Data Organisation

The embedded Equipment Issue Return report in the Dashboard showed five test records (TRX-1001 to TRX-1005) in the screenshot, with a record counter, search, sorting and an add button. All records in this report are test data entered for demonstration; no real personal data is used.

CHAPTER 4
REPORTS AND DASHBOARDS

4.1 Overview

Reports present the records of a form as a table or board that can be searched, sorted and exported. The Dashboard is a page that combines summary figures, charts, links and an embedded report. This chapter describes the 13 reports and the Dashboard in the exported application.

4.2 Reports Implemented

The application has seven list reports (one for each form) and six kanban reports, which group the records of a form by a status or type field. Table 4.1 lists them.

Table 4.1: Reports in the Application

S.No.	Report Name	Type	Source Form	Grouping / Feature
1	All Department Registrations	List	Department Registration	—
2	All Equipment Registrations	List	Equipment Registration	Totals row on Purchase Cost
3	Equipment Categories	Kanban	Equipment Registration	Equipment Category
4	All Borrow Requests	List	Borrow Request	—
5	Request Statuses	Kanban	Borrow Request	Request Status
6	All Approvals	List	Request Approval	Filter on Approval Date
7	Approval Statuses	Kanban	Request Approval	Approval Status
8	Equipment Issue Return Report	List	Equipment Issue Return	—
9	Return Statuses	Kanban	Equipment Issue Return	Return Status
10	All Damage Reports	List	Damage Report	Totals row on Estimated Repair Cost
11	Damage Types	Kanban	Damage Report	Damage Type
12	All Maintenance Requests	List	Maintenance Request	Totals row on Repair Cost
13	Maintenance Types	Kanban	Maintenance Request	Maintenance Type

The list reports of the equipment, damage and maintenance forms show a total, average, minimum and maximum row for their cost fields. The report menu offers Edit, Duplicate, Delete, Print, Add, Import and Export on the header and Edit, Duplicate, Delete and Print on each record.

4.3 Conditional Formatting

Six list reports colour the cells of one field when it takes a chosen value (purple, green or red). Table 4.2 lists the formatted field and values.

Table 4.2: Conditional Formatting Rules in the Reports

Report	Field and values highlighted
Equipment Issue Return Report	Return Status: Damaged, Not Issued, Returned
All Equipment Registrations	Equipment Condition: Damaged, Good, Under Maintenance
All Damage Reports	Damage Type: Electrical Failure, Missing Component, Software Issue
All Approvals	Approval Status: Approved, Rejected
All Borrow Requests	Request Status: Approved, Cancelled, Completed
All Maintenance Requests	Maintenance Type: Calibration, Other, Preventive Maintenance

4.4 Required Reports and Their Status

The project brief planned seven reports. Table 4.3 compares them with what was built.

Table 4.3: Planned Reports versus Reports Built

Planned Report	Report in the Application	Status
Equipment Inventory Report	All Equipment Registrations	Built
Available Equipment Report	Equipment Categories (grouped by category); no availability filter	Not built as a separate report
Borrowing Request Report	All Borrow Requests; Request Statuses	Built
Active Borrowing Report	Equipment Issue Return Report; Return Statuses	Covered by the Issued status; no dedicated filter
Overdue Equipment Report	Return Statuses (Overdue is a status value)	Not built as a separate report
Damage & Maintenance Report	All Damage Reports; All Maintenance Requests; Damage Types; Maintenance Types	Built as separate reports
Equipment Utilization Report	—	Not built (future enhancement)

4.5 Dashboard

The Dashboard is a page named “Dashboard” and is the first item of the application menu. Table 4.4 lists its components as defined in the page.

Table 4.4: Dashboard Components

Component	Definition in the page	Value in the screenshot
Tile 1 – Total Purchase Costs	Sum of Purchase Cost, Equipment Registration	12,000.00
Tile 2 – Total Repair Costs	Sum of Estimated Repair Cost, Damage Report	13,905.00
Tile 3 – Total Repair Costs	Sum of Repair Cost, Maintenance Request	14,220.49
Tile 4 – Count of Equipment Issue Return	Count of records, Equipment Issue Return	5
Chart 1 – Maintenance Type by Repair Cost	Bar chart on Maintenance Request; x = Maintenance Type, y = Repair Cost	Bars for Repair, Preventive Maintenance, Other, Calibration
Condition at Return panel	Four counts of Equipment Issue Return where Condition at Return is New, Fair, Good or Damaged	New 2, Fair 0, Good 0, Damaged 3
Quick Links	Four text links that open a new-record form in a new window: Equipment Issue Return, Equipment Registration, Damage Report, Department Registration	—
Chart 2 – Maintenance Status by Repair Cost	Bar chart on Maintenance Request; x = Maintenance Status, y = Repair Cost	Bars for Under Repair, Requested, Cancelled
Embedded report	Equipment Issue Return Report	5 records

Tiles 2 and 3 have the same label but are different measures: tile 2 adds the estimated repair costs entered in Damage Reports and tile 3 adds the actual repair costs entered in Maintenance Requests. The labels should be made distinct (see Section 8.4).

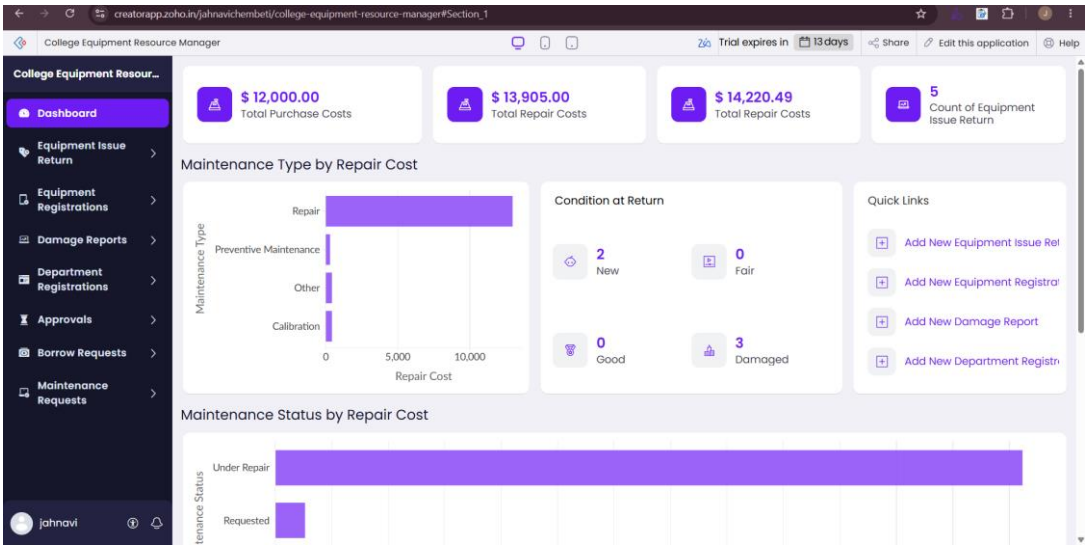


Figure 4.1: Dashboard Page: Summary Tiles, Charts and Quick Links

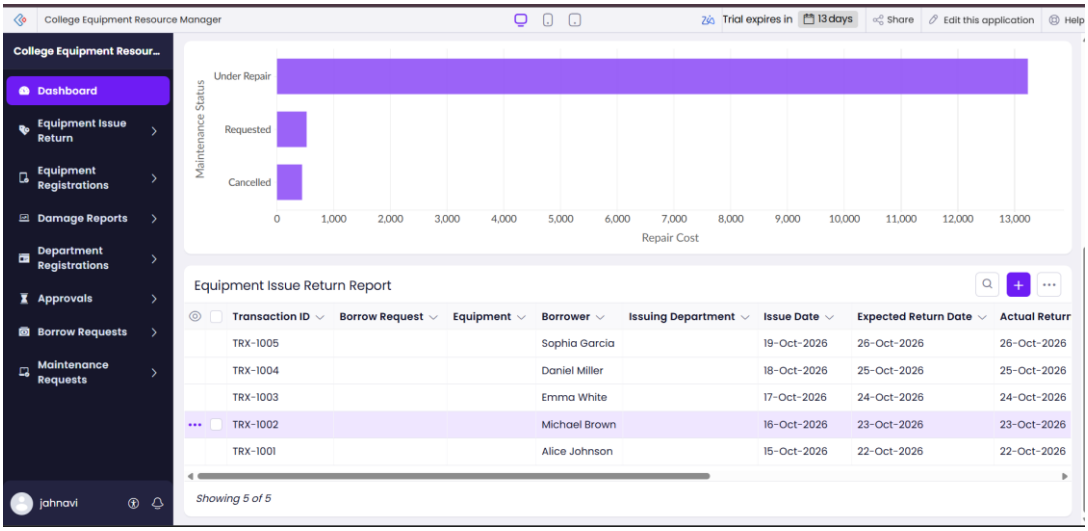


Figure 4.2: Dashboard Page: Maintenance Status Chart and Embedded Equipment Issue Return Report

4.6 Dashboard Indicators Planned but Not Built

The brief also named total equipment, available equipment, borrowed equipment, pending requests and overdue equipment as indicators. These are not on the Dashboard and are listed as future enhancements in Chapter 14. The Condition at Return panel and the maintenance charts cover the damaged and under-maintenance cases.

CHAPTER 5
PAGES (UI BUILDER)

5.1 Overview

The application has one page, the Dashboard. The Home Page, Equipment Marketplace and Department Dashboard of the project brief were not built; the Dashboard serves as the landing screen.

5.2 Landing Screen

The application opens on the Dashboard, the highlighted first menu item. It has no separate welcome page.

5.3 Navigation Menu

The menu is a dark side panel with the application name at the top and the logged-in user's name at the bottom. Each section has an arrow that expands it to show its forms and reports. Table 5.1 lists the sections.

Table 5.1: Application Menu Structure

Menu Item	Form	Reports of the form
Dashboard	—	Dashboard page
Equipment Issue Return	Equipment Issue Return	Equipment Issue Return Report; Return Statuses
Equipment Registrations	Equipment Registration	All Equipment Registrations; Equipment Categories
Damage Reports	Damage Report	All Damage Reports; Damage Types
Department Registrations	Department Registration	All Department Registrations
Approvals	Request Approval	All Approvals; Approval Statuses
Borrow Requests	Borrow Request	All Borrow Requests; Request Statuses
Maintenance Requests	Maintenance Request	All Maintenance Requests; Maintenance Types

5.4 Quick Links to Forms

The Dashboard's Quick Links panel has four links, each opening a new-record form in a new window: Equipment Issue Return, Equipment Registration, Damage Report and Department Registration. There are no quick links for Borrow Request, Request Approval or Maintenance Request.

5.5 Dashboard Components

The components are listed in Table 4.4. They are arranged in a grid: four summary tiles in the first row, then a chart, a condition panel and the quick links, then a second chart and the embedded report.

5.6 Reports and Charts

The Dashboard embeds the Equipment Issue Return Report so that recent transactions can be seen without leaving the page. The charts are described in Table 4.4.

5.7 Items Not Present

The Dashboard has no list of upcoming returns, overdue items or pending requests, and there is no searchable page of available equipment for borrowers. These are future enhancements.

5.8 Interface and Device Layouts

The exported definition holds separate layout settings for web, tablet and phone: labels are placed to the left on the web, a two-column layout is used for one device class and a sliding pane for the tablet. The Dashboard screenshots show desktop, tablet and phone preview icons in the header.

CHAPTER 6
WORKFLOWS AND AUTOMATION

6.1 Overview of Workflow Types

Zoho Creator offers form workflows, schedules, approvals, payments, blueprints, batch workflows and functions [1]. The application has five automations: two form workflows and one schedule workflow, which contain Deluge scripts, and two Blueprints, which contain stages and transitions but no scripts. Table 6.1 summarises them and Figure 6.1 shows the form workflows tab. The complete scripts are in Appendix B.

Table 6.1: Automation Summary

Automation	Type	Form	Trigger	Status
Validate Borrow Request Dates	Form workflow	Borrow Request	On add or edit; on validate	Enabled
Notify Manager on High Priority Request	Form workflow	Borrow Request	On add; on success	Enabled
Warranty Expiry Reminder	Schedule workflow	Equipment Registration	Date of Warranty Expiry, 10:00 (Asia/Kolkata)	Inactive in the export
Equipment Borrowing Workflow	Blueprint	Borrow Request	Starts at “Request Drafted”	13 stages, 25 transitions
Equipment Maintenance Lifecycle	Blueprint	Maintenance Request	Starts at “Request Created”	14 stages, 20 transitions

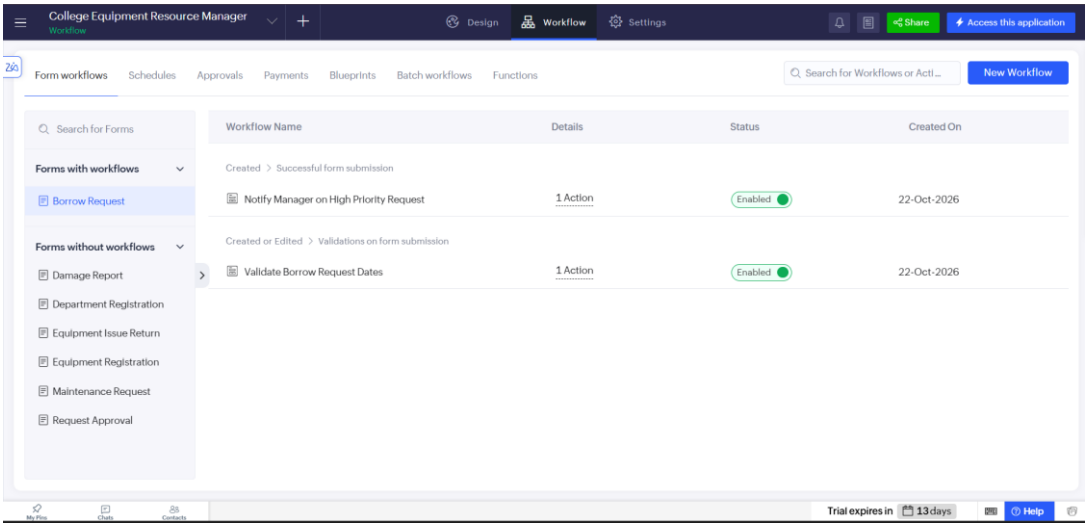


Figure 6.1: Form Workflows of the Borrow Request Form

6.2 Validate Borrow Request Dates

This form workflow runs when a Borrow Request is added or edited, in the on-validate stage, before the record is saved. If both Required From and Required Until are filled and Required Until is earlier than Required From, it shows the message “The return date (required_until) cannot be before the start date (required_from). Please correct the dates.” against the Required Until field and cancels the submission. Script B.1.

6.3 Notify Manager on High Priority Request

This form workflow runs after a Borrow Request has been added successfully. It fetches the saved record and, if the priority is “Urgent”, looks up the requesting department, takes the department’s Contact Email and sends an HTML e-mail with the subject “Urgent Equipment Request Notification – <Request ID>”. The e-mail gives the request ID, requester, department, purpose and the required-from and required-until times, and asks the manager to review and approve. The sender is the logged-in user. Script B.2.

The workflow’s name says “High Priority” but the script tests only the value “Urgent”. A request with priority High does not trigger the e-mail. This is recorded in Table 11.3.

6.4 Warranty Expiry Reminder

This schedule workflow is set on the Warranty Expiry field of Equipment Registration and starts at 10:00 in the Asia/Kolkata time zone. When it runs, it finds equipment whose warranty expires in the next 30 days, looks up each item’s department and, if the Department Head field is not empty, sends a plain-text e-mail “Warranty Expiry Alert: <equipment name>” with the equipment name, ID and expiry date. Script B.3.

Three points in the exported definition need attention. The workflow has the status “inactive”, so it does not run until it is activated. It sends the e-mail to the value of Department Head, which is a Single Line field and holds a name unless an e-mail address is typed in it. And the schedule starts on the expiry date itself while the script looks 30 days ahead; the message says “will expire in 30 days”. See Table 11.3.

6.5 Equipment Borrowing Workflow (Blueprint)

A Blueprint defines the stages of a record and the transitions allowed between them. The Equipment Borrowing Workflow is attached to the Borrow Request form. It starts at “Request Drafted” and has 13 stages: Request Drafted, Request Submitted, Pending Department Approval, Pending Lab Manager Approval, Equipment Reserved, Equipment Issued, Equipment In Use, Pending Return, Equipment Damaged, Equipment Lost, Equipment Returned, Inspection Pending and Request Closed. Table

6.2 lists its 25 transitions and Figure 6.2 shows them. None of the transitions has a before or after action in the exported definition.

Table 6.2: Transitions of the Equipment Borrowing Workflow

S.No.	Transition	From stage	To stage
1	Approve Department Request	Pending Department Approval	Pending Lab Manager Approval
2	Approve Equipment Request	Pending Lab Manager Approval	Equipment Reserved
3	Cancel Reservation	Equipment Reserved	Pending Lab Manager Approval
4	Close Request	Inspection Pending	Request Closed
5	Close Request with Damage Report	Equipment Damaged	Request Closed
6	Close Request with Loss Report	Equipment Lost	Request Closed
7	Confirm Receipt	Equipment Issued	Equipment In Use
8	Extend Borrow Period	Pending Return	Equipment In Use
9	Issue Equipment	Equipment Reserved	Equipment Issued
10	Mark as Due for Return	Equipment In Use	Pending Return
11	Process Return	Pending Return	Equipment Returned
12	Reject Department Request	Pending Department Approval	Request Drafted
13	Reject Lab Request	Pending Lab Manager Approval	Request Drafted
14	Report Damage	Equipment In Use	Equipment Damaged
15	Report Damage Found	Equipment Returned	Equipment Damaged
16	Report Loss	Equipment In Use	Equipment Lost
17	Request Department Re-evaluation	Pending Lab Manager Approval	Pending Department Approval
18	Request Repair and Return	Equipment Damaged	Pending Return
19	Request Replacement	Equipment Lost	Pending Return
20	Request Re-submission	Inspection Pending	Pending Return
21	Return for Revision	Request Submitted	Request Drafted
22	Return to Storage	Equipment Issued	Equipment Reserved
23	Route to Department Head	Request Submitted	Pending Department Approval
24	Submit for Inspection	Equipment Returned	Inspection Pending
25	Submit Request	Request Drafted	Request Submitted



Figure 6.2: Stage and Transition Diagram of the Equipment Borrowing Workflow (Drawn from the Blueprint Definition)

6.6 Equipment Maintenance Lifecycle (Blueprint)

The Equipment Maintenance Lifecycle is attached to the Maintenance Request form. It starts at “Request Created” and has 14 stages: Request Created, Request Under Review, Request Rejected, Request Approved, Request Awaiting Parts, Request Cancelled, Maintenance Scheduled, Maintenance In Progress, Maintenance On Hold, Maintenance Completed, Maintenance Reopened, Maintenance Cancelled, Maintenance Pending Approval and Maintenance Closed. Table 6.3 lists its 20 transitions and Figure 6.3 shows them. It also has no before or after actions.

Table 6.3: Transitions of the Equipment Maintenance Lifecycle

S.No.	Transition	From stage	To stage
1	Approve Completion	Maintenance Pending Approval	Maintenance Closed
2	Approve Request	Request Under Review	Request Approved
3	Cancel Maintenance	Maintenance On Hold	Maintenance Cancelled
4	Cancel Due to Parts Unavailability	Request Awaiting Parts	Request Cancelled
5	Cancel Request	Request Approved	Request Cancelled
6	Complete Additional Work	Maintenance Reopened	Maintenance Completed
7	Complete Maintenance	Maintenance In Progress	Maintenance Completed
8	Mark as Incomplete	Maintenance Completed	Maintenance Reopened
9	Parts Received	Request Awaiting Parts	Request Approved
10	Pause Maintenance	Maintenance Scheduled	Maintenance On Hold
11	Reject Completion	Maintenance Pending Approval	Maintenance Reopened
12	Reject Request	Request Created	Request Rejected
13	Reopen for Additional Work	Maintenance In Progress	Maintenance Reopened
14	Request Parts	Request Under Review	Request Awaiting Parts
15	Resume Maintenance	Maintenance On Hold	Maintenance In Progress
16	Resume Work	Maintenance Reopened	Maintenance In Progress
17	Schedule Maintenance	Request Approved	Maintenance Scheduled
18	Start Maintenance	Maintenance Scheduled	Maintenance In Progress
19	Submit for Approval	Maintenance Completed	Maintenance Pending Approval
20	Submit for Review	Request Created	Request Under Review

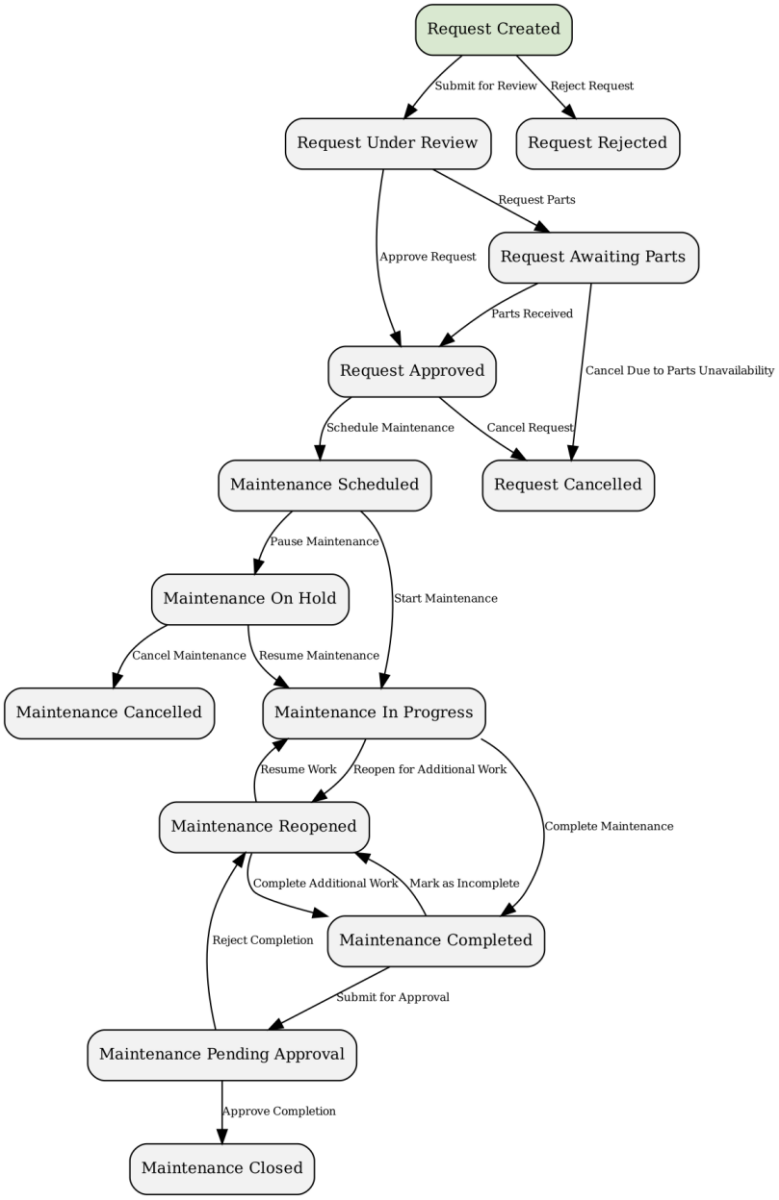


Figure 6.3: Stage and Transition Diagram of the Equipment Maintenance Lifecycle (Drawn from the Blueprint Definition)

6.7 Deluge Scripting Notes

Deluge is the scripting language of Zoho Creator [2]. Table 6.4 lists the three scripts of the application. The Blueprints contain no script.

Table 6.4: Deluge Scripts Used

Script	Workflow	Event	Deluge features used
B.1	Validate Borrow Request Dates	Borrow Request: on add or edit, on validate	input values, null check, comparison, alert, cancel submit

Script	Workflow	Event	Deluge features used
B.2	Notify Manager on High Priority Request	Borrow Request: on add, on success	fetch record by ID, condition, lookup navigation, string concatenation, sendmail (HTML)
B.3	Warranty Expiry Reminder	Equipment Registration: schedule on Warranty Expiry	date-range fetch (next 30 days), for each loop, isNull, sendmail

6.8 Testing of Workflows

Test cases TC-06 to TC-12 in Table 11.1 cover the workflows and Blueprints.

CHAPTER 7
INTEGRATION CAPABILITIES

7.1 Overview

The application uses no external service, API, webhook or connection. Its only outbound channel is e-mail sent by the Deluge sendmail task from two workflows. Other integration options of Zoho Creator [1] are listed as proposals.

7.2 Integrations in the Application

Table 7.1: Integration Options and Their Status

Option	Use in this project	Status
E-mail (sendmail)	Urgent-request notification; warranty-expiry reminder	Implemented in two workflows (Scripts B.2, B.3)
Import and export of data	Import and Export are on every list report's menu	Available on the reports; no import file is part of this report
Zoho Creator REST API	Read equipment records from another system	Proposed; not built
Webhooks	Send an event to another service when equipment is returned	Proposed; not built
Connections to other services	Calendar entries for return dates	Proposed; not built

CHAPTER 8
AI AND ANALYTICS

8.1 Overview

This chapter separates analytics, which the Dashboard provides, from artificial intelligence, which the application does not use. The exported definition contains no prediction model, AI field or AI-driven feature.

8.2 Analytics Provided by the Dashboard

The Dashboard sums purchase cost, estimated repair cost and repair cost, counts issue and return records, counts equipment by condition at return, and charts repair cost by maintenance type and by maintenance status. These are descriptive figures obtained by adding and counting records; they report what has happened and do not predict what will happen.

8.3 Decisions Supported

The repair-cost charts let a department compare spending on repair with preventive maintenance and calibration. A rising count of items returned as Damaged can point to items or borrowers that need attention. Because the figures in the screenshots come from test data, no conclusion about the college is drawn from them.

8.4 Dashboard Observation

Two tiles carry the label “Total Repair Costs”. The first (13,905.00) is the sum of Estimated Repair Cost in Damage Reports and the second (14,220.49) is the sum of Repair Cost in Maintenance Requests. They measure different things and the labels should be changed, for example to “Total Estimated Damage Cost” and “Total Maintenance Repair Cost”. The cost fields use US dollars, while the application time zone is America/Los_Angeles and the schedule runs in Asia/Kolkata; for a college in India the currency and time zone should be reviewed.

8.5 Possible AI Extensions (Future)

Possible future uses include predicting which equipment is likely to fail from its repair history and suggesting a maintenance date. These need much more data than a test dataset and are listed in Chapter 14.

8.6 Use of Zia and Other AI Assistance

Zoho Creator includes Zia App Builder, which can generate an application skeleton from a description [5].

CHAPTER 9
SYSTEM ARCHITECTURE

9.1 Architectural Overview

The application follows the standard architecture of a Zoho Creator application: a browser-based interface, an application layer of forms, reports, a page and workflows, and a data layer managed by the platform. There is no separate server code to deploy; the platform hosts and runs the application. The application is reached at the address given in Appendix C.

9.2 Layers

Table 9.1: Architecture Layers

Layer	Components in this project
Presentation	Menu, Dashboard page (tiles, two charts, quick links, embedded report), 7 forms and 13 reports shown in a web browser
Application logic	2 form workflows and 1 schedule workflow (3 Deluge scripts); 2 Blueprints
Data	7 forms and 75 fields, joined by 11 lookups
Security	Zoho Creator profiles: Read, Write, Developer, Administrator and Customer; one role, CEO
Communication	E-mail sent by the sendmail task
Platform	Zoho Creator cloud (India data centre, creatorapp.zoho.in)

9.3 Data Flow

A borrower submits a Borrow Request; the date workflow checks the dates and, if the priority is Urgent, the notification workflow e-mails the department. The Blueprint moves the request through drafting, submission, department and laboratory approval, reservation, issue, use, return and inspection. A manager's decision is recorded in Request Approval; the handing over and return are recorded in Equipment Issue Return; damage and maintenance records are added when needed; and the Dashboard summarises the result.

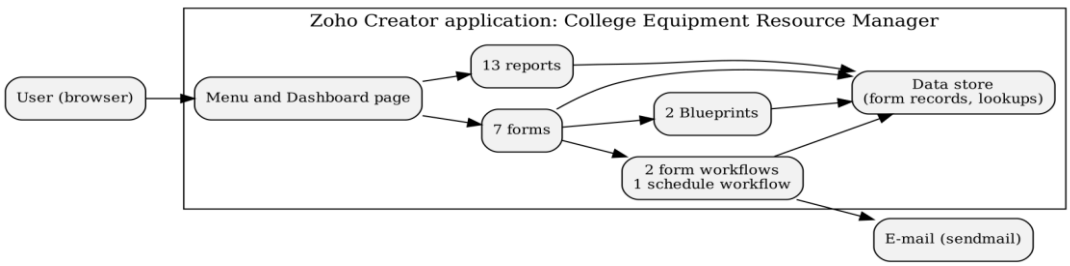


Figure 9.1: Architecture of the College Equipment Resource Manager

CHAPTER 10
IMPLEMENTATION

10.1 Implementation Approach

The application was built in the order in which the parts depend on one another: master forms first, transaction forms next, then reports, the Dashboard, workflows and Blueprints. Table 10.1 states each step and the evidence for it.

Table 10.1: Implementation Steps and Evidence

Step	Activity	Result	Evidence
1	Create the application	College Equipment Resource Manager	Export header; menu
2	Create the seven forms and their 75 fields	Done	Tables 3.2–3.8
3	Join the forms by lookups	11 lookups	Table 3.9; Figure 3.2
4	Create reports	7 list and 6 kanban reports	Table 4.1
5	Build the Dashboard page	Tiles, 2 charts, quick links, embedded report	Table 4.4; Figures 4.1, 4.2
6	Add form and schedule workflows	2 form workflows, 1 schedule (inactive)	Table 6.1; Appendix B
7	Add Blueprints	2 Blueprints (25 and 20 transitions)	Tables 6.2, 6.3
8	Enter test data	Five issue and return records seen	Figure 4.2
9	Test the application	Test cases defined; execution is recorded at demonstration	Chapter 11
10	Share with users	Default profiles only; no custom profile	Chapter 12

10.2 Creating the Application

The application was created in Zoho Creator under the account jahnavichembeti with the link name college-equipment-resource-manager, date format dd-MMM-yyyy and 24-hour time. Its export was generated on 08-Nov-2026 (version 1.0).

10.3 Creating Forms and Fields

The seven forms were created with the fields of Chapter 3. Identifier fields were made Single Line text; choices were entered as drop-down values with an “Other” option; costs are Currency in US dollars.

10.4 Configuring Lookups

Lookup fields were configured as drop-downs filled from the parent form, with the parent's identifier shown and an "Add New" option, as in Table 3.9.

10.5 Creating Reports

A list report was created for each form and a kanban report for the six forms that have a status or type field. Conditional formatting was added to six list reports and totals to the cost fields of three.

10.6 Building the Dashboard

The Dashboard page was built from text tiles bound to form data, two bar charts, a group of count tiles filtered by condition, four quick links and an embedded report (Table 4.4).

10.7 Creating Workflows and Blueprints

The two form workflows and the schedule were written in Deluge (Appendix B). The two Blueprints were defined with their stages and transitions (Tables 6.2 and 6.3).

10.8 Profiles and Sharing

The application carries the five default profiles of Zoho Creator and the default CEO role. No custom profile or permission set is present (Chapter 12).

10.9 Entering Test Data

Test records such as TRX-1001 to TRX-1005 were entered to check the reports and Dashboard. They are invented demonstration data.

10.10 Deployment

The application runs in live mode on the Zoho Creator account of the student at https://creatorapp.zoho.in/jahnavichembeti/college-equipment-resource-manager. The link opens only for users who have been given access to the account's application. It has not been published to the college's users, and no production deployment is claimed.

CHAPTER 11
TESTING AND RESULTS

11.1 Testing Strategy

The test cases cover positive, negative, validation, workflow, Blueprint and unauthorised-access scenarios for the forms, workflows, Dashboard and profiles. For each case, Table 11.1 gives the step, the expected result and the basis in the exported application. The last two columns are completed by hand during the demonstration, with Pass or Fail and any remark. No case has been recorded as passed in this document.

11.2 Test Cases

Table 11.1: Test Cases

ID	Area	Type	Test Step	Expected Result	Basis in the Export	Result	Remark
TC-01	Equipment Registration	Positive	Add an equipment record with all fields filled	Record saved; success message “Equipment Registration Added Successfully”; shown in All Equipment Registrations	Form and report exist		
TC-02	Equipment Registration	Negative	Submit the form with every field empty	Result depends on field settings	Export shows no mandatory field; record is expected to save (observation O-06)		
TC-03	Department Registration	Positive	Add a department with an e-mail and status Active	Saved; available in lookups	Form and lookups exist		
TC-04	Equipment Registration	Validation	Enter the same Serial Number twice	Second record refused if unique is set	Export shows no unique setting		
TC-05	Borrow Request	Positive	Add a request with valid dates and priority Normal	Saved; no e-mail sent	Script B.2 sends e-mail only for Urgent		
TC-06	Borrow Request	Validation	Set Required Until earlier than Required From and submit (Validate Borrow Request Dates)	Submission cancelled; message shown against Required Until	Script B.1: alert and cancel submit		
TC-07	Borrow Request	Validation	Edit a saved request so that Required Until is earlier than Required From	Edit refused with the same message	Event is “on add or edit”		
TC-08	Borrow Request	Workflow	Add a request with priority Urgent and a department with a valid Contact Email	HTML e-mail sent to the department’s Contact Email	Script B.2		
TC-09	Borrow Request	Workflow	Add a request with priority High	No e-mail is sent (script tests Urgent only)	Script B.2; observation O-01		

ID	Area	Type	Test Step	Expected Result	Basis in the Export	Result	Remark
TC-10	Blueprint	Positive	Open a new request and perform Submit Request, then Route to Department Head	Stage changes Request Drafted → Request Submitted → Pending Department Approval	Table 6.2		
TC-11	Blueprint	Positive	Take a maintenance request through Submit for Review, Approve Request, Schedule Maintenance, Start Maintenance, Complete Maintenance	Stage follows the transitions	Table 6.3		
TC-12	Warranty reminder	Workflow	Activate the schedule and run it with equipment expiring within 30 days and a Department Head e-mail	E-mail “Warranty Expiry Alert” sent	Script B.3; schedule is inactive (O-02)		
TC-13	Request Approval	Positive	Add an approval with status Approved and a date	Saved; row highlighted in All Approvals	Conditional format on Approved		
TC-14	Request Approval	Negative	Reject a request leaving Rejection Reason empty	Result depends on field settings	No mandatory field in export		
TC-15	Equipment Issue Return	Positive	Add a transaction linked to a borrow request and an equipment item	Saved; count on Dashboard rises by one	Lookups; Dashboard tile 4		
TC-16	Equipment Issue Return	Positive	Edit the record with Actual Return Date, Condition at Return Good and Return Status Returned	Saved; Condition at Return panel changes	Dashboard panel		
TC-17	Damage Report	Positive	Add a report linked to equipment and a transaction	Saved; added to Estimated Repair Cost total	Tile 2		
TC-18	Maintenanc e Request	Positive	Add a request with type Repair and a cost	Saved; chart and tile 3 change by that cost	Tile 3; Charts 1 and 2		
TC-19	Dashboard	Positive	Compare tiles 2 and 3 with the totals shown in the two cost reports	Each tile equals its report total	Table 4.4		
TC-20	Access control	Unauthori sed	Open the application as a user with the Read profile and try to add a record	Add refused	Profile permissions per Zoho Creator		
TC-21	Access control	Unauthori sed	Open the application link without signing in	Sign-in page shown; no data visible	Platform authentication		

11.3 Test Summary

Table 11.2: Test Summary

Type	Cases planned
Positive	11
Negative	2
Validation	3
Workflow	3
Unauthorised	2
Total	21

11.4 Observations from the Review of the Exported Definition

Reading the exported application produced the observations in Table 11.3. They are not test failures; they are points where the definition does not match the intention stated in the workflow name or the report text, and each has a recommended correction.

Table 11.3: Observations on the Application

ID	Observation	Recommended correction
O-01	Notify Manager on High Priority Request tests priority = “Urgent” only; High does not trigger it.	Test for High or Urgent, or rename the workflow.
O-02	Warranty Expiry Reminder has the status inactive.	Activate the schedule before use.
O-03	The warranty reminder sends mail to Department Head, a text field; the notification workflow uses Contact Email.	Use Contact Email in the warranty script.
O-04	The schedule starts on the expiry date but the script looks 30 days ahead and says “will expire in 30 days”.	Start the schedule daily, or start it 30 days before expiry.
O-05	Two Dashboard tiles are both labelled Total Repair Costs but measure estimated damage cost and actual maintenance cost.	Rename the tiles (Section 8.4).
O-06	No field is mandatory or unique and identifiers are typed text.	Set mandatory fields; use Auto Number for identifiers.
O-07	Only the default profiles exist; the planned roles are not defined.	Create profiles for each role (Chapter 12).
O-08	Costs are in US dollars and the application time zone is America/Los_Angeles.	Use Indian rupees and Asia/Kolkata.
O-09	Borrow Request has no field naming the equipment requested.	Add a lookup or subform to Equipment Registration.

11.5 Scenario Screenshots

The screenshots available for this report are the builder view (Figure 3.1), the Dashboard (Figures 4.1 and 4.2) and the form workflows (Figure 6.1). Screenshots of the test scenarios are taken while the test cases are executed and attached to Appendix C.

CHAPTER 12
SECURITY AND DATA PRIVACY

12.1 Access Control

Zoho Creator controls access through profiles and roles [1]. The exported application contains the five default profiles – Read, Write, Developer, Administrator and Customer – and the default role CEO, which can access the data of all other users. No custom profile for the planned user types (Table 2.1) has been created, so separation between administrators, managers, faculty, technicians and students is not yet enforced by the application.

Table 12.1: Profiles in the Application

Profile	Type	Description in the export
Read	Users' permissions	Read permission for all components
Write	Users' permissions	Write permission for all components
Developer	Developer	Developer profile
Administrator	Users' permissions	All permissions
Customer	Customer portal	Default profile with add and view permission only

12.2 Data Privacy

Borrower names, e-mail addresses and purposes are personal data. Only test data is used in this report. Before real use, the college should decide who may see borrower details and how long records are kept.

12.3 Security Measures

Table 12.2: Security Considerations

Area	Measure	Status
Authentication	Sign-in with a Zoho account	Provided by the platform
Authorisation	Profiles and roles	Default profiles only (O-07)
Input validation	Date check on Borrow Request	Implemented (Script B.1)
Record integrity	Mandatory and unique fields	Not configured (O-06)
E-mail content	Notification contains request details	Sent to the department's contact e-mail only

12.4 Risks

The main risks are broad permissions (every user with the Write profile can change every form) and personal data left in a shared account. Both are addressed by creating role-specific profiles and by tests TC-20 and TC-21.

CHAPTER 13
ADVANTAGES AND LIMITATIONS

13.1 Advantages

- One application holds equipment, requests, approvals, issue and return, damage and maintenance, replacing separate registers.
- The Dashboard gives totals and charts without a separate reporting tool.
- Low-code configuration shortened development; most of the application was built by configuration, with three short Deluge scripts [3], [4].
- The Blueprints give borrowing and maintenance a defined sequence of stages.
- The data model is relational: details are entered once and linked by lookup.

13.2 Limitations

- No mandatory, unique or default-value settings; identifiers are typed by the user.
- Only default profiles exist, so role-based separation is not enforced.
- Borrow Request does not record which equipment is requested.
- The warranty reminder is inactive and has the timing and recipient issues of Section 6.4.
- The Blueprints define stages but carry no actions, and their stage is not linked to the status fields.
- The Dashboard lacks planned indicators (available, borrowed and overdue equipment, pending requests).
- The data are test data; behaviour with real volumes is unknown.

13.3 Comparison with the Existing Manual System

Compared with registers and spreadsheets, the application gives a single view of equipment, requests and costs and sends e-mail for urgent requests. The size of the benefit has not been measured.

CHAPTER 14
FUTURE ENHANCEMENTS

All items below are future work.

Table 14.1: Future Enhancements

Enhancement	Description	Benefit
Role-based profiles	Create profiles for administrator, manager, faculty, technician and student, with form and report permissions	Controlled access
Corrections of Table 11.3	Urgent and High notification, active warranty schedule with correct recipient and timing, distinct tile names	Behaviour matches intention
Mandatory fields and Auto Number	Make key fields mandatory and generate identifiers	Cleaner data
Equipment in the request	Add a lookup or subform for the equipment requested	Request tied to equipment
Blueprint actions	Update status fields and send mail on transitions	Less manual updating
Overdue alerts	Daily schedule that e-mails borrowers and managers about overdue items	Fewer unreturned items
More Dashboard indicators	Total, available, borrowed, overdue equipment and pending requests	Fuller overview
Equipment marketplace page	Searchable list of available equipment	Self-service discovery
Barcode / QR codes	Scan to issue or return	Faster, fewer errors
Predictive maintenance	Use repair history to suggest maintenance dates	Less downtime

CHAPTER 15
CONCLUSION

This report describes the College Equipment Resource Manager, a Zoho Creator application that manages equipment records, borrowing requests, approvals, issue and return, damage reports and maintenance in a college. The application has seven forms with 75 fields joined by eleven lookups, thirteen reports, a Dashboard page, two form workflows, one schedule workflow and two Blueprints with 25 and 20 transitions.

The review of the exported definition also showed what remains to be done: role-specific profiles, mandatory fields, an equipment field on the borrow request, activation and correction of the warranty reminder, and the Dashboard indicators of the brief. These are recorded in Table 11.3 and Chapter 14, and the test cases of Chapter 11 are to be executed and recorded at the demonstration.

Within these limits, the work shows how a low-code platform can cover the main data, process and reporting needs of equipment management mostly by configuration, with only three short scripts.

CHAPTER 16
REFERENCES

[1] Zoho Corporation, “Zoho Creator Help,” Zoho Creator Knowledge Base. [Online]. Available: https://help.zoho.com/portal/en/kb/creator

[2] Zoho Corporation, “Deluge Help,” Zoho Deluge Documentation. [Online]. Available: https://www.zoho.com/deluge/help/

[3] T. Bock and U. Frank, “Low-Code Platform,” Business & Information Systems Engineering, vol. 63, no. 6, pp. 733–740, 2021.

[4] D. Di Ruscio, D. Kolovos, J. de Lara, A. Pierantonio, M. Tisi, and M. Wimmer, “Low-code development and model-driven engineering: Two sides of the same coin?,” Software and Systems Modeling, vol. 21, no. 2, pp. 437–446, 2022.

[5] Zoho Corporation, “Creating applications with Zia App Builder,” Zoho Creator Help. [Online]. Available: https://help.zoho.com/portal/en/kb/creator/developer-guide/applications/create-applications/articles/creating-applications-with-zia-app-builder

APPENDIX A – IMPLEMENTATION EVIDENCE CHECKLIST

"Export" means the application definition (.ds file) generated on 08-Nov-2026. "Screenshot" means a screenshot supplied for this report.

Item	Found	Evidence
Application created	Yes	Export; menu header
Seven forms	Yes	Export; Figure 3.1
75 fields with types and choices	Yes	Export; Tables 3.2–3.8
11 lookups	Yes	Export; Table 3.9
Subform	No subform in the application	Export
13 reports (7 list, 6 kanban)	Yes	Export; Table 4.1
Dashboard page	Yes	Export; Figures 4.1, 4.2
2 form workflows	Yes	Export; Figure 6.1
Schedule workflow	Yes (inactive)	Export
2 Blueprints	Yes	Export; Tables 6.2, 6.3
Deluge scripts	3	Appendix B
Custom roles and profiles	No (defaults only)	Export
Integrations other than e-mail	No	Export
Test data	Yes (5 issue records)	Figure 4.2
Test execution record	To be completed at demonstration	Table 11.1
Production deployment	Not deployed to college users	Section 10.10

APPENDIX B – DELUGE SOURCE CODE

The scripts below are reproduced from the application's exported definition.

B.1 Validate Borrow Request Dates (Borrow Request – on add or edit, on validate)

```
// Validate that required until date is not before required from da
te
if(input.required until != null && input.required from != null)
{
    if(input.required until < input.required from)
    {
        alert(required until,"The return date (required until) cann
ot be before the start date (required from). Please correct the dat
es.");
        cancel submit;
    }
}
```

B.2 Notify Manager on High Priority Request (Borrow Request – on add, on success)

```
// Fetch the current borrow request record
borrowRequest = borrow request[ID == input.ID];
// Check if the request priority is Urgent
if(borrowRequest.priority == "Urgent")
{
    // Get the requesting department details
    departmentRecord = department registration[ID == borrowRequest.
requesting department.ID];
    // Get the department head email
    departmentHeadEmail = departmentRecord.contact email;
    // Prepare notification message
    notificationMessage = "<p>Dear Manager,</p>" + "<p>An urgent eq
uipment request has been submitted.</p>" + "<p><strong>Request ID:<
/strong> " + borrowRequest.request id + "</p>" + "<p><strong>Reques
ter:</strong> " + borrowRequest.requester name + "</p>" + "<p><stro
ng>Department:</strong> " + departmentRecord.department name + "</p
>" + "<p><strong>Purpose:</strong> " + borrowRequest.purpose + "</p
>" + "<p><strong>Required From:</strong> " + borrowRequest.required
 from.toString() + "</p>" + "<p><strong>Required Until:</strong> "
+ borrowRequest.required until.toString() + "</p>" + "<p>Please rev
iew and approve this request at your earliest convenience.</p>";
    // Send notification email to department head
    sendmail
    [
        from :zoho.loginuserid
        to :departmentHeadEmail
        subject :"Urgent Equipment Request Notification - " + borro
wRequest.request id
        message :notificationMessage
        content type :HTML
    ]
}
```

B.3 Warranty Expiry Reminder (Equipment Registration – schedule)

```
// Fetch equipment records where warranty expires within the next 3
0 days
warrantyExpiringRecords = equipment registration[warranty expiry in
 next 30 days];
// Iterate through each record and send email to department head
for each  equipment in warrantyExpiringRecords
{
    // Get the department record using the lookup field
    departmentRecord = department registration[ID == equipment.depa
rtment.ID];
    // Check if department head email is available
    if(departmentRecord.department head.isNull() == false)
    {
        // Send email to department head
        sendmail
        [
            from :zoho.loginuserid
            to :departmentRecord.department head
            subject :"Warranty Expiry Alert: " + equipment.equipmen
t name
            message :"Dear Department Head,\n\nThis is to inform yo
u that the warranty for the following equipment will expire in 30 d
ays.\n\nEquipment Name: " + equipment.equipment name + "\nEquipment
 ID: " + equipment.equipment id + "\nWarranty Expiry Date: " + equi
pment.warranty expiry.toString() + "\n\nPlease take necessary actio
n to renew the warranty or replace the equipment.\n\nRegards,\nEqui
pment Management Team"
        ]
    }
}
```

The two Blueprints contain no Deluge code. Their stages and transitions are listed in Tables 6.2 and 6.3.

APPENDIX C – LINKS AND SCREENSHOT CHECKLIST

C.1 Links

Application (live mode)	https://creatorapp.zoho.in/jahnavichembeti/college-equipment-resource-manager
Zoho Creator account	jahnavichembeti (India data centre, creatorapp.zoho.in)
Zoho Creator Help [1]	https://help.zoho.com/portal/en/kb/creator
Deluge Help [2]	https://www.zoho.com/deluge/help/

The application link opens only for users who have been given access to the account's application.

C.2 Screenshots

No.	Screenshot	Included
1	Builder search view (forms, reports, workflows, page)	Yes – Figure 3.1
2	Dashboard – tiles, charts and quick links	Yes – Figure 4.1
3	Dashboard – status chart and embedded report	Yes – Figure 4.2
4	Form workflows of Borrow Request	Yes – Figure 6.1
5	ER diagram, Blueprint diagrams, architecture diagram	Yes – drawn from the export (Figures 3.2, 6.2, 6.3, 9.1)
6	Test scenario screenshots taken during execution of Table 11.1	To be added at demonstration

APPENDIX D – USER MANUAL

The steps below follow the forms and reports of the application. Because only the default profiles exist, every signed-in user with the Write or Administrator profile can perform all steps.

D.1 Opening the Application

1. Sign in to Zoho Creator with the account given by the administrator.
2. Open College Equipment Resource Manager from the link in Appendix C.
3. The Dashboard opens first.

D.2 Registering Departments and Equipment

1. Use Add New Department Registration on the Dashboard (or the Department Registrations menu) and fill the department ID, name, head, contact e-mail, number, building, room and status.
2. Use Add New Equipment Registration and fill the equipment details; choose the department from the lookup.

D.3 Submitting a Borrow Request

1. Open Borrow Requests and choose Add.
2. Enter the request ID, requesting department, requester name and e-mail, purpose, required from and until, priority and request status.
3. Submit. If Required Until is earlier than Required From, the form shows a message and does not save. If the priority is Urgent, an e-mail is sent to the department’s contact e-mail.

D.4 Moving a Request through the Blueprint

1. Open the request and use the transition button shown for its current stage, for example Submit Request, Route to Department Head or Approve Department Request.
2. Continue until the request reaches Request Closed.

D.5 Recording an Approval

1. Open Approvals and choose Add.
2. Choose the borrow request, enter the manager, status, date and comments; add a rejection reason if rejected.

D.6 Issuing and Returning Equipment

1. Use Add New Equipment Issue Return on the Dashboard.

2. Choose the borrow request, equipment and issuing department; enter the borrower, issue date, expected return date, condition at issue and issued by.
3. On return, edit the record and enter the actual return date, condition at return, returned to and return status.

D.7 Reporting Damage and Maintenance

1. Use Add New Damage Report and link the equipment and the transaction; give the type, severity, estimated cost and status.
2. Open Maintenance Requests and choose Add; link the equipment and damage report, give the type, technician, dates, repair cost and status. Use the Blueprint buttons to move the request through its stages.

D.8 Reading the Dashboard and Reports

The tiles show total purchase cost, total estimated damage cost, total repair cost and the number of issue and return records. The charts show repair cost by maintenance type and by status. The Condition at Return panel counts items by condition. Open any report from the menu to search, sort, print, import or export.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name		Register No.	
Project Title	Hearing & Client Tracker for Small Law Practices	Date	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner **Signature of External Examiner**

Name: _____ Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name		Register No.	
Department / Section		Date of Viva	
Application URL Verified	Yes / No	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER ***EXTERNAL EXAMINER***

Name: _____ Name: _____